AI COMIC STORY STUDIO

C!

ComicCraft

AI Comic Story Creator Using Gemini Models

Turns a single story idea into a structured, illustrated comic book with live web preview and downloadable A4 PDF export.

Python 3.11+ | FastAPI | Google Gemini AI | Pydantic v2 | SDXL / Diffusers | fpdf2 PDF Export

HOW COMICCRAFT WORKS

01 **Plan Story**
Outline & Character Bible

02 **Write Script**
Captions & Dialogues

03 **Draw Panels**
AI or Placeholder Artwork

04 **Export PDF**
Web Preview & A4 Book

Introduction & Abstract

What ComicCraft is and how it creates a complete comic from one short idea

PROJECT ABSTRACT

- **Simple User Input:** User gives a short story idea, character name, setting, tone, art style, and panel count (3 to 8 panels).
- **Two-Stage Gemini AI:** First plans the story outline and Character Bible, then writes panel captions, narration, and dialogues.
- **Validated JSON Data:** Uses Pydantic v2 to check all AI outputs so panel count and order are always accurate.
- **Panel Artwork & PDF:** Generates one PNG image per panel and exports a clean A4 Comic Book PDF using fpdf2.
- **Offline Ready:** Includes a built-in Demo Mode and Placeholder Art so the full app works even without API keys.

FOUR-STEP PIPELINE

01 — PLAN

Creates comic title, Character Bible, and panel scenes.

02 — WRITE

Writes panel captions, narration, and character dialogues.

03 — ILLUSTRATE

Generates one consistent PNG image for every panel.

04 — PUBLISH

Shows responsive web preview and builds an A4 PDF.

Problem Statement & Objectives

Challenges in comic creation and how ComicCraft solves them

PROBLEM STATEMENT

1. Needs Multiple Skills

Creating a comic manually needs story writing, drawing, and page layout skills.

2. Unstructured AI Text

Normal AI chat replies return plain text, which breaks easily when splitting into panels.

3. Inconsistent Characters

Independent image prompts make the hero's face and outfit change in every panel.

4. API & Key Failures

Rate limits, model changes, or missing API keys stop normal AI apps from working.

PROJECT OBJECTIVES

1. One-Stop Comic Studio

Convert one story brief into a complete 3–8 panel comic via Web UI or JSON API.

2. Strict JSON Validation

Use Gemini JSON mode and Pydantic schemas so panel structure never breaks.

3. Character Continuity

Keep the hero visually consistent across all panels using a Character Bible.

4. Always-Working Fallback

Support automatic retries and an Offline Demo Mode that needs no API keys.

Existing System vs Proposed System

How ComicCraft improves upon manual methods and basic AI scripts

Feature	Existing System	Proposed System (ComicCraft)
Workflow	Separate tools for writing, drawing, and layout.	**Single web app from story prompt to A4 PDF.**
AI Output	Plain text output that breaks parsers easily.	**Structured JSON output validated by Pydantic v2.**
Character Look	Character face and clothes change every panel.	**Fixed Character Bible repeated in every panel.**
Fault Tolerance	Crashes on API errors or without internet.	**Auto-retry + fallback models + Offline Demo Mode.**
Output & Save	Manual PDF arrangement; files get overwritten.	**Auto A4 PDF + saved in unique UUID folders.**

Core Advantage: Combines AI story planning, consistent character prompts, and automatic PDF creation in one place.

Technology Stack & Requirements

Tools, libraries, and basic system specifications used to build ComicCraft

BACKEND

Python 3.11+ · FastAPI · Uvicorn

Handles web routes, REST JSON endpoints, and interactive API docs (/docs).

VALIDATION

Pydantic v2 · .env Settings

Validates form inputs, AI JSON structure, and protects API keys safely.

AI STORY

Google Gemini API (google-genai)

Generates structured JSON story outlines and full panel scripts in two stages.

PANEL ARTWORK

Pillow · Hugging Face · Diffusers

Creates local Placeholder PNGs, Cloud SDXL images, or Local Stable Diffusion art.

FRONTEND UI

HTML5 · CSS3 · JS · Jinja2

Clean responsive creation form, loading overlay, and comic panel preview.

PDF & STORAGE

fpdf2 · DejaVu Fonts · UUID Storage

Exports Unicode A4 PDFs and saves each comic in its own local UUID folder.

MINIMUM SYSTEM REQUIREMENTS

Python 3.11+ • Modern Web Browser • 4 GB RAM • 250 MB Disk Space • No GPU or External Database Required

System Architecture

Clean modular flow from user input to web preview and A4 PDF file

Web Browser UI

Story Form & Comic Preview

REST JSON API

POST /api/v1/comics

Health & Docs

GET /health • Swagger /docs

▼

FASTAPI BACKEND & PYDANTIC INPUT VALIDATION

Validates Story Brief (3–8 Panels), Routes Requests & Handles Errors Cleanly

▼

COMIC SERVICE (MAIN PIPELINE CONTROLLER)

Coordinates Story -> Panel Artwork -> PDF Export -> Local Storage

▼

Story Service

llm.py

- 2-Stage Gemini AI
- Outline + Script JSON
- Offline Demo Writer

Image Service

images.py

- Pillow Placeholder
- Hugging Face SDXL
- Local Stable Diffusion

PDF Service

pdf_service.py

- fpdf2 A4 Builder
- Styled Cover Page
- One Page per Panel

Storage Service

repository.py

- Unique UUID Folder
- Panel PNG Images
- comic.pdf & comic.json

Two-Stage Gemini AI Workflow

How ComicCraft plans the story first and keeps characters visually consistent

STEP 1

Story Brief

User Input

- Story idea
- Main character
- Setting & tone
- Art style
- 3 to 8 panels

→

STEP 2

Stage 1: Outline

Gemini Outline Model

- Returns strict JSON
- Creates Comic Title
- Builds Character Bible
- Plans N panel scenes

→

STEP 3

Stage 2: Script

Gemini Story Model

- Uses approved outline
- Writes panel captions
- Adds narration & dialogue
- Builds image prompts

→

STEP 4

Panel Artwork

Image Provider

- Repeats hero traits
- Deterministic seed
- 1 PNG per panel
- Sent to Preview & PDF

WHY TWO STAGES?

Planning the outline in Stage 1 and writing dialogue in Stage 2 gives better story pacing and guarantees the exact number of panels.

WHAT IS THE CHARACTER BIBLE?

A fixed visual description of the hero's face, hairstyle, and outfit created in Stage 1 and repeated in every panel's image prompt.

Key Features & Smart Fallback Design

Built for reliable execution both online with AI models and offline in Demo Mode

1. SMART AI FALLBACKS

- **Auto Retry:** Automatically retries temporary API errors (429 / 503) using exponential backoff.
- **Backup Models:** Switches immediately to fallback Gemini models if a primary model is unavailable.
- **Offline Demo Mode:** Generates complete demo stories offline when no Gemini API key is configured.

2. FLEXIBLE IMAGE MODES

- **Three Providers:** Supports Pillow Placeholder, Hugging Face Cloud SDXL, and Local Stable Diffusion.
- **Per-Panel Protection:** If one panel image fails, only that panel uses a placeholder—the comic still finishes.
- **Clear Badges:** Shows honest provider labels and fallback warnings right on the preview page.

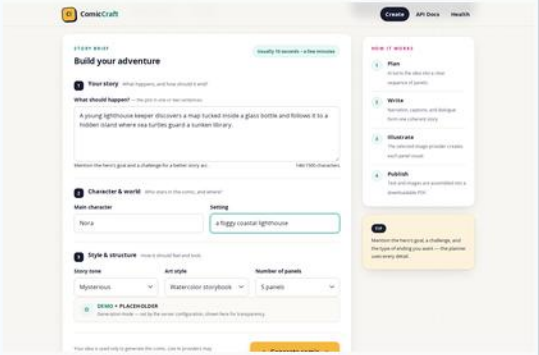
3. PDF & SAFE STORAGE

- **A4 PDF Export:** Builds a styled cover page and per-panel pages with Unicode DejaVu fonts.
- **UUID Workspaces:** Stores every comic safely in its own isolated folder (PNGs, PDF, and JSON).
- **Reopen Anytime:** Saved comics can be viewed at /comics/{id} or downloaded at /download/{id}.

Application Output & Screenshots

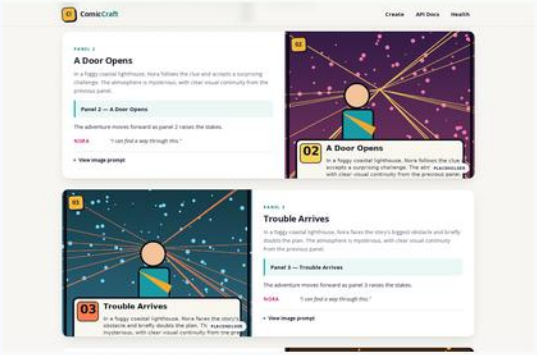
Real screens from the ComicCraft web studio and exported A4 PDF

1. Comic Creation Form



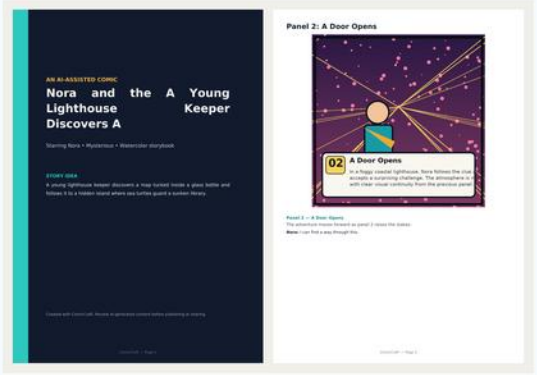
Clean web form to enter story idea, character, setting, tone, art style, and panel count.

2. Generated Panels Preview



Responsive preview pairing each panel's artwork with its caption, narration, and dialogue.

3. Exported A4 Comic PDF



Downloadable A4 PDF book featuring a styled dark cover page and one page per panel.

Testing, Future Scope & Conclusion

Verified quality checks, upcoming features, and project summary

TESTING & QUALITY

- **11 Automated Tests:** All 11 pytest test cases pass completely offline with ~80% code coverage.
- **Verified Routes:** Tested web creation form, REST JSON API, PDF export, and error handling.
- **OpenAPI Docs:** Interactive Swagger UI at /docs allows direct testing of JSON API routes.

FUTURE ENHANCEMENTS

- **Background Queue:** Add async job queues with live progress bars for heavy AI image models.
- **Cloud Accounts:** Add user login and cloud storage for saving personal comic libraries.
- **More Creative Control:** Single-panel regeneration, multi-panel page grids, and Tamil language support.

CONCLUSION

- **End-to-End Solution:** Turns a simple story idea into an illustrated web comic and A4 PDF in minutes.
- **Consistent & Structured:** Uses two-stage Gemini JSON, Pydantic validation, and a Character Bible.
- **Reliable by Design:** Works smoothly with cloud AI models as well as 100% offline in Demo Mode.

Thank You!

ComicCraft • AI Comic Story Creator